GROUP WORK PROJECT # 03
GROUP NUMBER: 6782

MScFE 600: FINANCIAL DATA

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- Additional writing aids: Anti-Plagiarism Guide, Academic Writing Guide, Online Writing Resources.
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- Submission format tips:
  - Use the same font type and size and same format throughout your report. You can use Calibri 11, Arial 10, or Times 11.
  - Do NOT split charts, graphs, and tables between two separate pages.
  - Always include the axes labels and scales in your graphs as well as an explanation of how the data should be read.
- Use the LIRN Library for your research. It can be accessed via the left navigation pane inside the WQU learning platform.

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Step 1: Fallout Risks

Scenario	Model Failures	Financial Crises
Money at a fixed rate for an unsecured purchase (e.g. credit card) for an individual	1. Credit score models may fail to accurately predict default rates or similar risk assessments, leading to higher than expected defaults. 2. Financial models are usually based on various assumptions like macroeconomic, mathematical, statistical etc. These various assumptions serve as inputs for the model. If these assumptions happen to be inaccurate then the model may fail or lead to bad outcomes for the company. The fixed rate charge may turn out to be inadequate.	1. In a financial crisis, a lot of potential customers may lose jobs, reducing the number of signups. Additionally, it may add financial stress onto existing customers or borrowers making it more difficult for them to pay back their loans and thereby increasing default rates. 2. When there is a financial crisis, prices of assets go down and may lead to the borrower's inability to pay their debts. This will have adverse effects on the company. Given that the loan is unsecured, the company won't be able to recoup the payment from taking over the assets and selling them.
Money at a floating rate for a secured purchase (e.g. home or automobile) for	1. Models can underestimate the impact of rising interest rates on	1. A housing crash can lead to widespread defaults and foreclosures which

an individual	borrower affordability and default risk.	would reduce the value of the underlying assets or collateral. This can cause major losses for lenders as the market value of the collateral could be significantly lower than the outstanding loan balance.
Money at a fixed rate for a business for a construction loan	1. Inaccurate construction and project models may be inaccurate, leading to cost overruns and delays.	1. Reduced demand, liquidity issues, and inability for prime and subprime vendors to complete projects can lead to defaults or the project losing value. 2. A financial crisis can also negatively impact the viability of the project or business venture which the construction was meant for. This can lead to lower than the original estimated demand and customer base, thereby making it difficult for the borrower to repay the loan.
Publicly traded Equity (e.g. common stock)	1. Valuation models might fail to account for market conditions or company-specific risks, leading to overvaluation or undervaluation.	1. A financial crisis can lead to a market crash, where stock prices plummet, causing significant losses for investors and potentially leading to bankruptcies and layoffs in the affected companies which can lead to losses at their creditors and investors.
Publicly traded bond (e.g. treasury bond, corporate bond)	1. Inaccurate interest rate risk models can lead to incorrect duration and convexity estimates leading to inaccurate	1. Rising default rates for corporate bonds and government fiscal issues can lead to bond price declines and increased

	representations.	credit spreads, resulting in significant losses for bondholders.
An illiquid security – you choose the security	1. Valuation models for illiquid assets may be based on outdated or overly optimistic assumptions, leading to incorrect pricing. For example, in newly formed companies with limited financials and high growth opportunities, investors may be overly optimistic on the prospects and thus overvalue the assets.	1. During a financial crisis, the lack of marketability can prevent the sale of illiquid assets, leading to severe mark-to-market losses and potential insolvency for holders who need to maintain high liquidity.

Step 2: Identifying Data

Below are 12 types of data which can potentially be used:

Equity Prices:

- Data Type: Asset
- Data Processing: Raw Prices, Returns
- Data Frequency: Daily, Weekly, Quarterly
- Data Class: Equity
- Data Source: Exchanges, Vendors; ex Yahoo Finance,Charles schwab
- Data Variety: Actual Data, Adjusted Data

Bond Yields:

- Data Type: Asset
- Data Processing: Raw Yields, Implied Volatilities
- Data Frequency: Daily, Weekly, Quarterly
- Data Class: Fixed Income
- Data Source: Exchanges, OTC, Vendors; ex Yahoo Finance
- Data Variety: Observed Data, Adjusted Data

GDP Growth Rates:

- Data Type: Economic

- Data Processing: Levels
- Data Frequency: Quarterly, Annually
- Data Class: Economic
- Data Source: Government Agencies, Analytics Companies; ex FRED Economic data
- Data Variety: Observed Data

Unemployment Rates:

- Data Type: Economic
- Data Processing: Levels
- Data Frequency: Monthly
- Data Class: Economic
- Data Source: Government Agencies, Vendors; ex FRED Economic data
- Data Variety: Observed Data

Credit Ratings:

- Data Type: Ratings
- Data Processing: Levels/Categories
- Data Frequency: Annually, Quarterly
- Data Class: Credit
- Data Source: Ratings Agencies, Vendors; ex Experian
- Data Variety: Actual Data

Corporate Earnings:

- Data Type: Accounting
- Data Processing: Levels
- Data Frequency: Quarterly, Annually
- Data Class: Equity
- Data Source: Audited Financials, Vendors; ex 10-k reports
- Data Variety: Observed Data

Exchange Rates:

- Data Type: Asset
- Data Processing: Raw Prices, Returns
- Data Frequency: Daily, Weekly
- Data Class: FX
- Data Source: Exchanges, Vendors; ex Yahoo Finance
- Data Variety: Actual Data

Commodity Prices:

- Data Type: Asset
- Data Processing: Raw Prices, Returns
- Data Frequency: Daily, Weekly
- Data Class: Commodity
- Data Source: Exchanges, Vendors; ex: Yahoo finance
- Data Variety: Actual Data

Option Implied Volatilities:

- Data Type: Asset
- Data Processing: Implied Volatilities
- Data Frequency: Daily
- Data Class: Derivatives
- Data Source: Exchanges, Vendors
- Data Variety: Modeled Data

Real Estate Prices:

- Data Type: Asset
- Data Processing: Raw Prices, Returns
- Data Frequency: Quarterly, Annually
- Data Class: Real Estate
- Data Source: Government Agencies, Vendors; ex US Census
- Data Variety: Actual Data, Adjusted Data

Default Rates:

- Data Type: Credit
- Data Processing: Levels
- Data Frequency: Quarterly, Annually
- Data Class: Credit
- Data Source: Ratings Agencies, Analytics Companies; ex Moody's
- Data Variety: Observed Data

Interest Rates:

- Data Type: Economic
- Data Processing: Levels
- Data Frequency: Daily, Weekly, Quarterly
- Data Class: Economic
- Data Source: Government Agencies, Vendors; ex FRED Economic data
- Data Variety: Observed Data, Modeled Data

Step 3: Ethical Considerations Based on Fallout

Scenario	Ethical Challenges	Undesirable Result
1) Money at a fixed rate for an unsecured purchase (e.g., credit card) for an individual	1. Lenders may push credit cards to individuals who may not manage debt responsibly, prioritizing immediate profit over customer well-being and long term sustainability.	1. Individuals may accumulate unsustainable debt, leading to financial distress and higher default rates, ultimately affecting the lender.
2) Money at a floating rate for a secured purchase (e.g., home or automobile) for an individual	1. Borrowers may not fully understand the risks of floating rates, leading to potential financial strain if rates rise. Lenders might not provide clear information on the downsides of floating rates, especially in a low rate environment.	1. Borrowers could default on loans when interest rates increase, leading to foreclosures and repossessions, contributing to significant losses for the lender and economic distress.
3) Money at a fixed rate for a business for a construction loan	1. Lenders might push loans to businesses without thoroughly vetting them through due diligence, driven by the potential immediate profits from interest and fees.	1. Businesses may fail to complete projects due to unforeseen financial issues, leading to unfinished constructions, and financial losses which can affect all parties.
4) Publicly traded Equity (e.g., common stock)	1. Analysts and firms might provide overly optimistic projections to drive up stock prices for immediate gain.	1. Investors may suffer significant losses when the true financial health of companies is revealed, leading to loss of trust in financial markets, the company itself, and the reputational hit may

		not recover..
5) Publicly traded bond (e.g., treasury bond, corporate bond)	1. Credit rating agencies might inflate ratings due to conflicts of interest between them and their customers who may take their business elsewhere. This may mislead investors about the true risk of bonds.	1. Investors might face unexpected defaults and losses, leading to lower confidence in credit ratings and financial markets, and potential instability in the bond market which can cause the firm losses.
6) An illiquid security (e.g., private equity investment)	1. Fund managers might misrepresent the liquidity and risk of investments to attract more capital, prioritizing immediate fund growth over long term investor security. This could happen with overvalued startups or SMBs.	1. Investors could have trouble liquidating their positions, leading to financial losses and potential market distortions due to panic selling.

Step 4: Ethical Considerations Based on Data

Scenario 1: Money at a fixed rate for an unsecured purchase (e.g., credit card) for an individual

Ethical Challenges:

1. **Data Quality Problems**: Inaccurate or incomplete credit history data can lead to incorrect assessments of an individual's creditworthiness.
2. **Collecting Data without Permission**: Gathering data on individuals' financial behaviors and spending habits without their explicit consent.
3. **Failing to Disclose Fees or Conflicts of Interest**: Not fully informing customers about the terms and fees associated with their credit cards.

Best Practices:

- **Data Quality Assurance**: Implement rigorous data validation and cleaning processes to ensure the accuracy and completeness of credit history data.

- **Informed Consent**: Ensure that individuals are fully aware of what data is being collected and how it will be used. Obtain explicit consent before collecting personal financial data.
- **Transparency in Fees**: Clearly disclose all fees, interest rates, and terms associated with credit cards. Provide easy-to-understand summaries and use plain language in communications.

Scenario 2: Money at a floating rate for a secured purchase (e.g., home or automobile) for an individual

Ethical Challenges:

1. **Hiding Data**: Concealing information about the potential risks of floating rates, such as how much rates could increase.
2. **Extrapolating Data**: Making overly optimistic projections about future interest rates and property values without proper justification.
3. **Failing to Disclose Fees or Conflicts of Interest**: Not fully disclosing fees or the potential impact of rate changes on monthly payments.
4. **Manipulation of benchmark rates**: Lenders offering loans with floating rates sometimes manipulate benchmark rates for their own financial benefits. This can negatively impact the borrower with a floating rate loan.

Best Practices:

- **Full Disclosure**: Provide comprehensive information about the risks associated with floating rate loans, including historical rate fluctuations and potential future scenarios.
- **Justified Projections**: Ensure that all projections and forecasts are based on robust data and sound methodologies. Avoid making overly optimistic or unsubstantiated claims.
- **Transparency in Fees**: Clearly outline all associated fees and how rate changes can impact monthly payments. Provide tools and resources for borrowers to understand these impacts.

Scenario 4: Publicly traded Equity (e.g., common stock)

Ethical Challenges:

1. **Intentionally Creating Misleading Data**: Manipulating financial reports or earnings data to present a more favorable picture of a company's financial health.
2. **Propagating Fake Reviews**: Using false or misleading analyst reports to influence stock prices.
3. **Hiding Data**: Withholding negative information that could impact investors' decisions.

Best Practices:

- **Integrity in Reporting**: Adhere strictly to accounting standards and ethical guidelines to ensure that all financial reports accurately reflect the company's financial health.
- **Ethical Analyst Reports**: Ensure that all analyst reports and reviews are based on accurate and verified data. Avoid conflicts of interest where possible and maintain independence in analysis.
- **Transparency and Disclosure**: Commit to full transparency by disclosing all relevant information, both positive and negative. Regularly update investors on relevant changes in the company's performance or outlook.

Step 5: How to manage the risk of lending money at a fixed rate for an unsecured purchase

Scenario 1: Money at a fixed rate for an unsecured purchase (e.g., credit card) for an individual

This loan does not require collateral and also the rate of interest is fixed.

Unsecured loans are riskier for lenders than secured loans because there is no collateral to seize if the borrower defaults. So, the risk of default is higher. In order to mitigate this risk, lenders usually grant unsecured loans to only borrowers with high credit scores.

I'll be recommending the following methods to manage the risks:

I. **Credit Scores**: Loans should only be given to borrowers with high credit scores. Borrowers with high credit scores are less likely to default.
II. **Require a Co-signer**: A co-signer agrees to fulfill the debt if the borrower defaults. So, the co-signer is legally obliged to pay the outstanding loan balance if the borrower should default. This can also allow the company to give loans to prospective borrowers with relatively lower credit scores. The borrower can also obtain more favorable loan terms from the company, especially if the co-signer credit score is very high.
III. **Use of a collection Agency**: The company can also hire a collection agency to help to recover the debts if the borrower defaults. The collection agency will get a percentage of the amount of the outstanding loan balance recovered. The company can also negotiate with the Collection Agency to purchase the debt for a fee lower than the outstanding loan balance. The agency will then attempt to recover the full loan balance from the borrower for their own benefits.

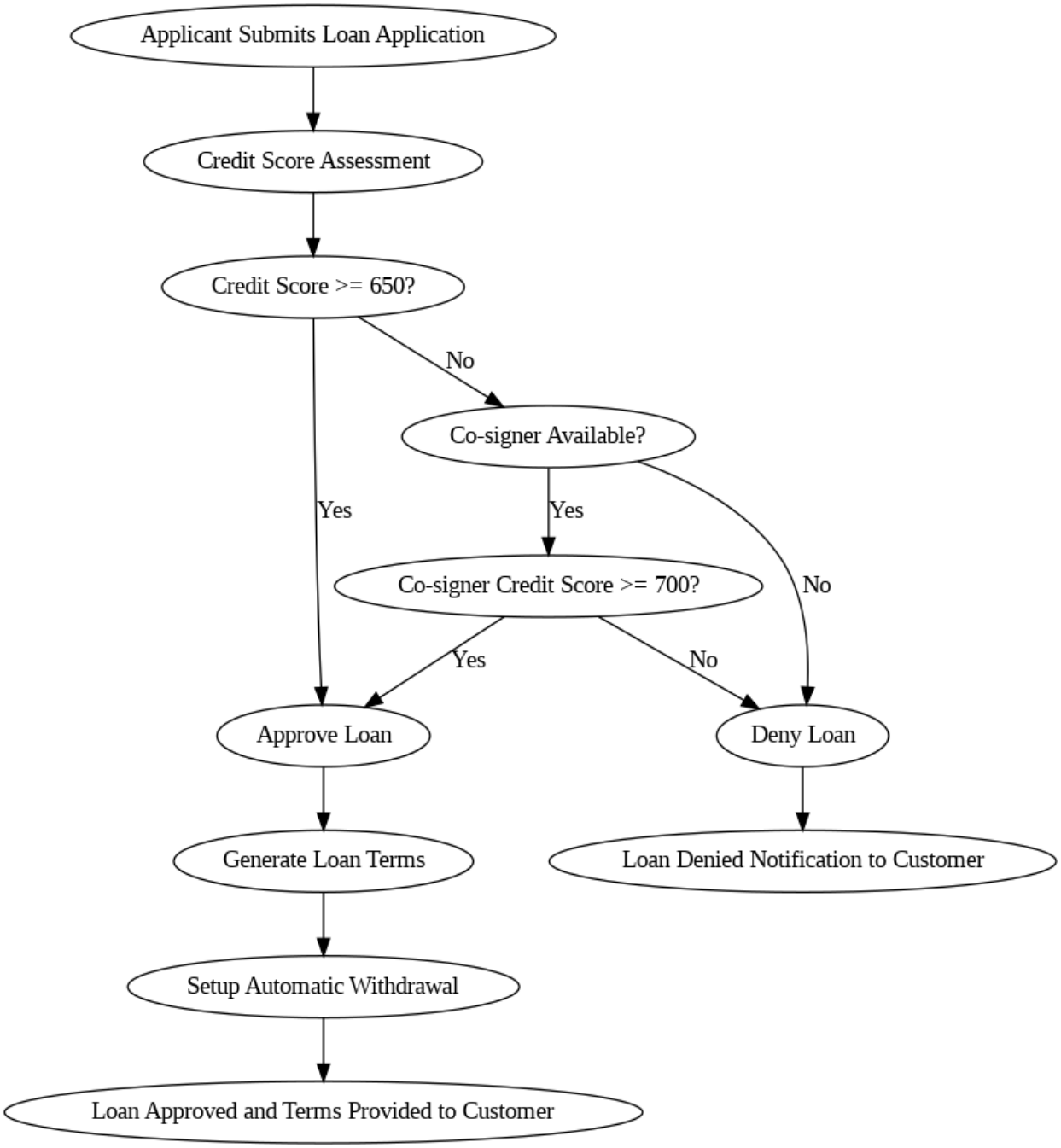
IV. **Legal Action**: The company may also consider taking the borrower to court should the borrower default.

V. **Automatic withdrawal from Borrower's account**: The company can also require that the borrowers accept automatic deductions from their bank accounts to repay the loan. The automatic withdrawal may be set around the day of the month that the borrower receives salary.

VI. **Diversification of the portfolio**: The company should also avoid concentrating loans to a single region or a particular sector of the economy. This will ensure that in case a particular region or sector of the economy is negatively impacted, the whole portfolio would not be affected adversely.

VII. **Continuous Review of loan Policy**: The company should ensure that lending policies are regularly reviewed to ensure that they are up to date with the current market trends and regulatory requirements. This would help to identify areas that require improvement and how they can be improved.

Given bellow the workflow diagram for lending money at a fixed rate for an unsecured purchase:



Step 6: Risk Management for Lending Money with Security

Risk Management in Lending with Publicly Traded Equity as Collateral

When lending money with publicly traded equity as collateral, it is essential to manage various risks effectively to protect the interests of both the lender and the borrower. The primary risks involved include market value volatility of the collateral, potential market downturns, and liquidity issues. Here's an in-depth overview of how these risks are managed in the lending process.

I. **Evaluation of Security Value:** The initial step in risk management is to accurately assess the market value of the security offered as collateral. This is accomplished by utilizing real-time market data to determine the current closing price of the security. For example, if an applicant provides shares of Apple Inc. (AAPL) as collateral, the lender retrieves the latest market price to appraise the value of the collateral.

II. **Loan-to-Value Ratio (LTV):** An essential aspect of risk management is to maintain an appropriate loan-to-value ratio. The LTV ratio is calculated by dividing the loan amount by the market value of the collateral. Typically, lenders establish a maximum LTV ratio (e.g., 50-70%) to ensure there is a buffer against market fluctuations. For example, if the collateral is valued at $100,000 and the lender sets a maximum LTV ratio of 70%, the maximum loan amount would be $70,000.

III. **Margin Calls**: To address the risk of market volatility, lenders may implement margin calls. A margin call occurs when the value of the collateral falls below a certain threshold, prompting the borrower to either deposit additional collateral or repay part of the loan to restore the LTV ratio to an acceptable level. This mechanism helps to protect the lender against significant declines in the value of the collateral.

IV. **Diversification**: Lenders can mitigate risk by diversifying the types of collateral they accept. By accepting a variety of publicly traded equities, lenders can reduce the impact of adverse price movements in any single security. This diversification strategy helps to spread risk across different assets and sectors.

V. **Continuous Monitoring**: Continuous monitoring of the collateral's market value is essential for effective risk management. Lenders should use automated systems to track the value of the collateral in real-time and issue alerts when the value approaches critical thresholds. This allows for prompt action to be taken, such as issuing margin calls or adjusting the loan terms.

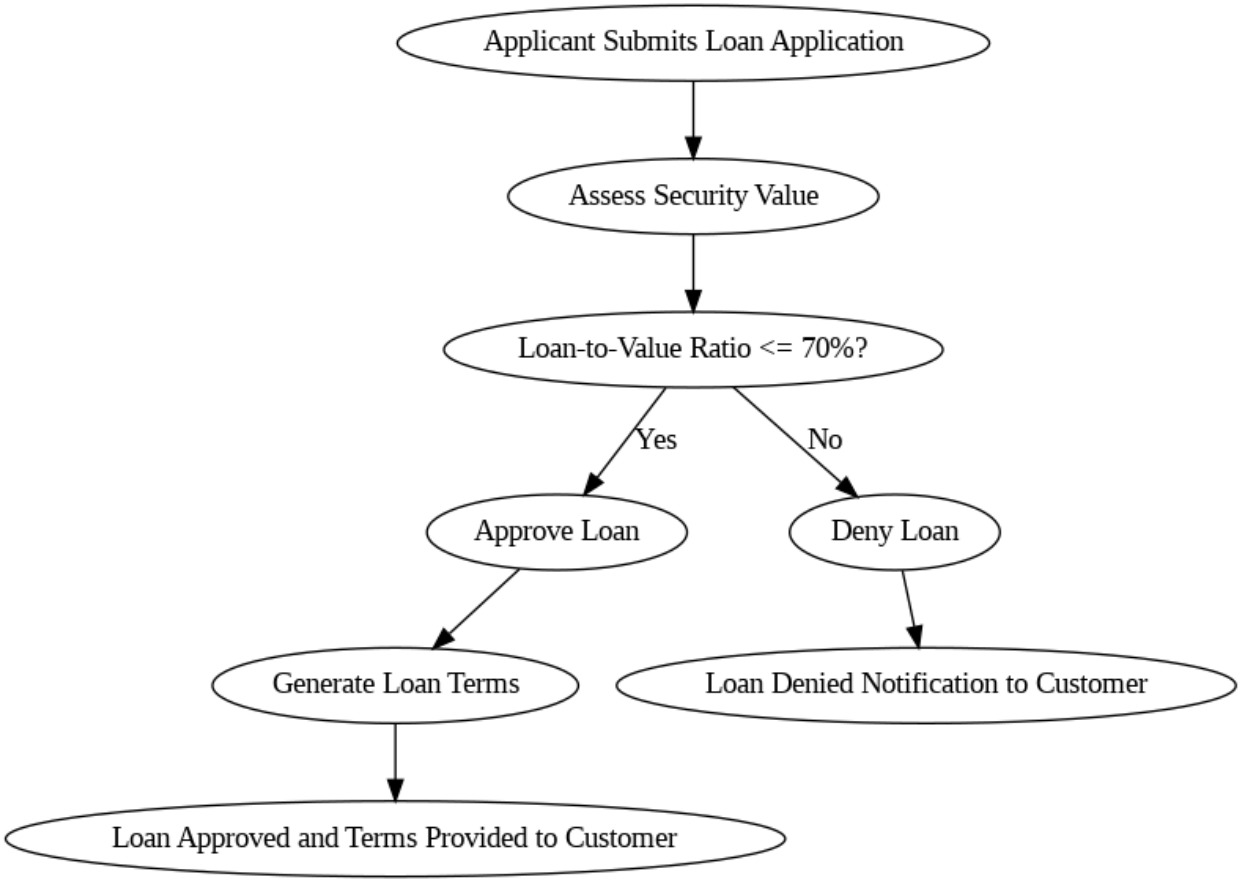
VI. **Hedging Strategies**: Lenders can employ hedging strategies to protect against downside risk. For example, using options or other derivatives, lenders can hedge against potential declines in the value of the collateral. This provides an additional layer of security and helps to stabilize the lender's financial position.

VII. **Legal and Regulatory Compliance**: Ensuring compliance with relevant legal and regulatory requirements is crucial for managing risk in lending with publicly traded equity as collateral. Lenders must adhere to securities laws, margin requirements, and other regulatory guidelines to protect both parties and maintain the integrity of the lending process.

VIII. **Borrower Assessment**: In addition to evaluating the collateral, lenders should also assess the creditworthiness of the borrower. This involves reviewing the borrower's financial history,

credit score, and repayment capacity. A thorough assessment helps to identify potential risks associated with the borrower's ability to repay the loan.

Given bellow the Workflow Diagram for Lending Money with Security



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